

# Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions.





# Project Overview

1

## Analyze Transactions

3,900 purchases across categories.

2

## Uncover Insights

Spending patterns, segments,  
preferences.

3

## Guide Decisions

Optimize strategy with data.

# Dataset Summary

Rows: 3,900

Columns: 18

- Customer demographics
- Purchase details
- Shopping behavior

Missing Data: 37 values in Review Rating.



# Exploratory Data Analysis (Python)

01

## Data Loading & Exploration

Imported with pandas, checked structure and statistics.

02

## Missing Data Handling

Imputed Review Rating using median by category.

03

## Column Standardization

Renamed to snake\_case for readability.

04

## Feature Engineering

Created age\_group and purchase\_frequency\_days.

05

## Database Integration

Loaded cleaned data to PostgreSQL for SQL analysis.



# Data Analysis (SQL)

1

Revenue by Gender

Female: \$75,191 | Male: \$157,890

2

High-Spending Discount Users

839 customers spent above average with discounts.

3

Top 5 Products by Rating

Gloves(3.86), Sandals(3.84), Boots(3.82), Hat(3.80), Skirt(3.78).

4

Shipping Type Comparison

Standard: \$58.46 | Express: \$60.48



# SQL Insights: Subscriptions & Discounts

## Subscribers vs. Non-Subscribers

|     |      |         |           |
|-----|------|---------|-----------|
| Yes | 1053 | \$59.49 | \$62,645  |
| No  | 2847 | \$59.87 | \$170,436 |

Non-subscribers contribute more revenue.

## Discount-Dependent Products

|          |        |
|----------|--------|
| Hat      | 50.00% |
| Sneakers | 49.66% |
| Coat     | 49.07% |
| Sweater  | 48.17% |
| Pants    | 47.37% |

Top 5 products with highest discounted purchases.

# Customer Segmentation & Top Products

## Customer Segments

Loyal

3116 Customers

Returning

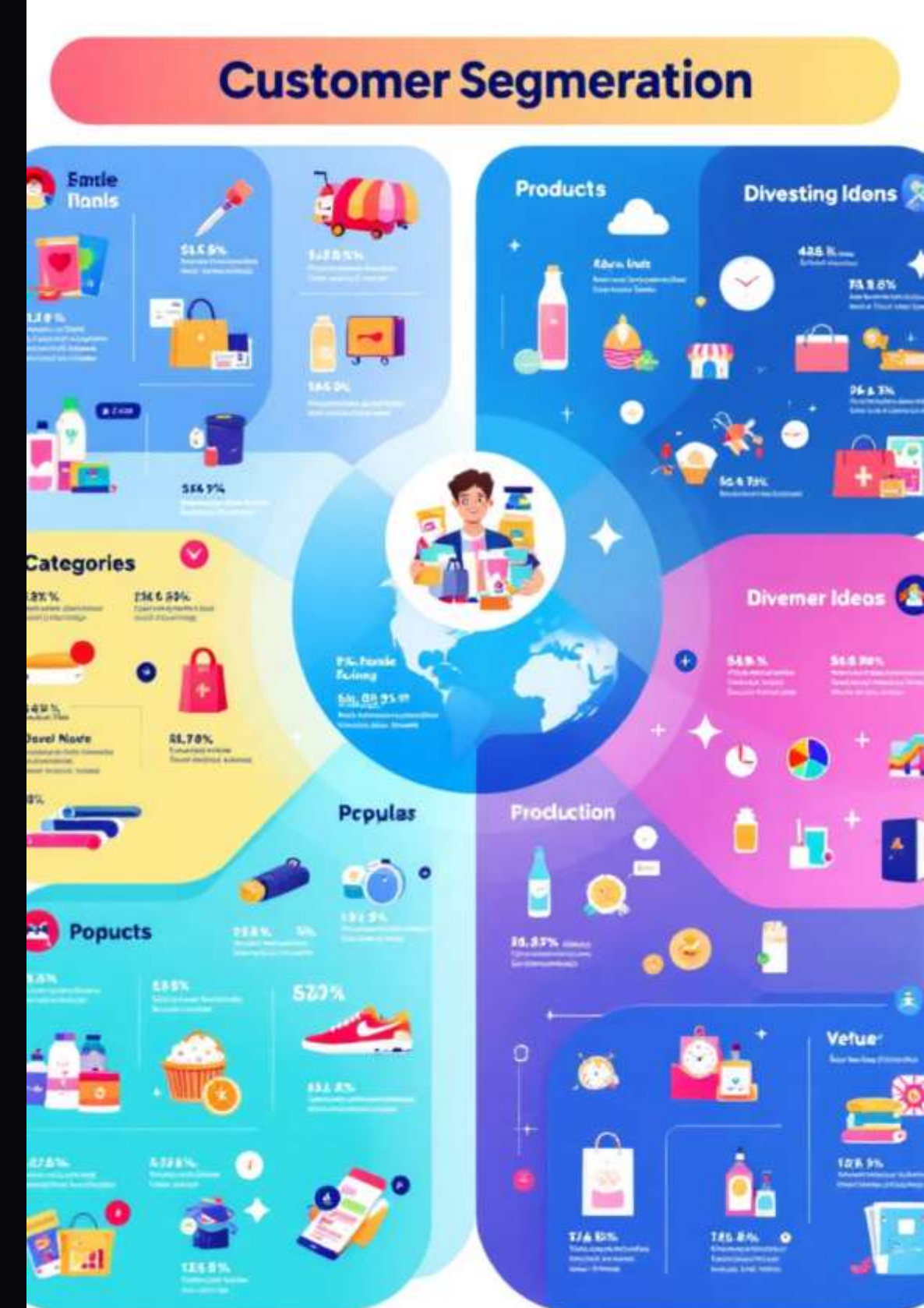
701 Customers

New

83 Customers

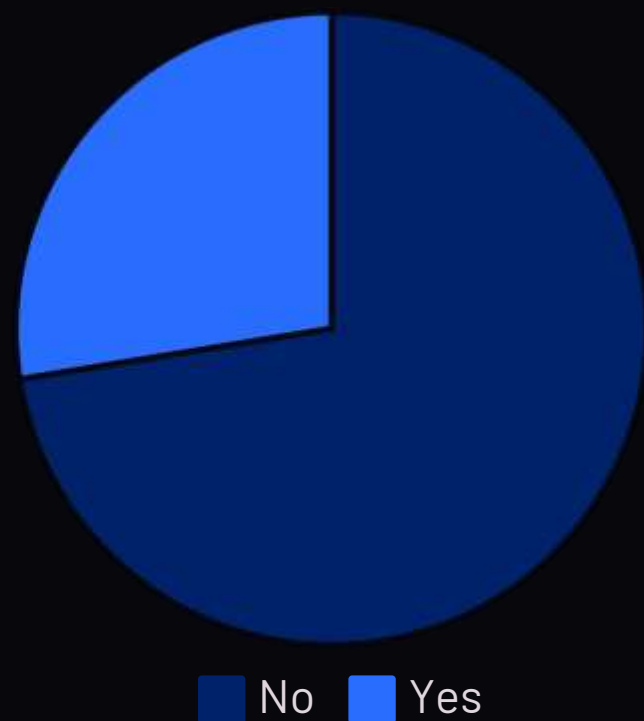
## Top Products per Category

- **Accessories:** Jewelry, Sunglasses, Belt
- **Clothing:** Blouse, Pants, Shirt
- **Footwear:** Sandals, Shoes, Sneakers
- **Outerwear:** Jacket, Coat



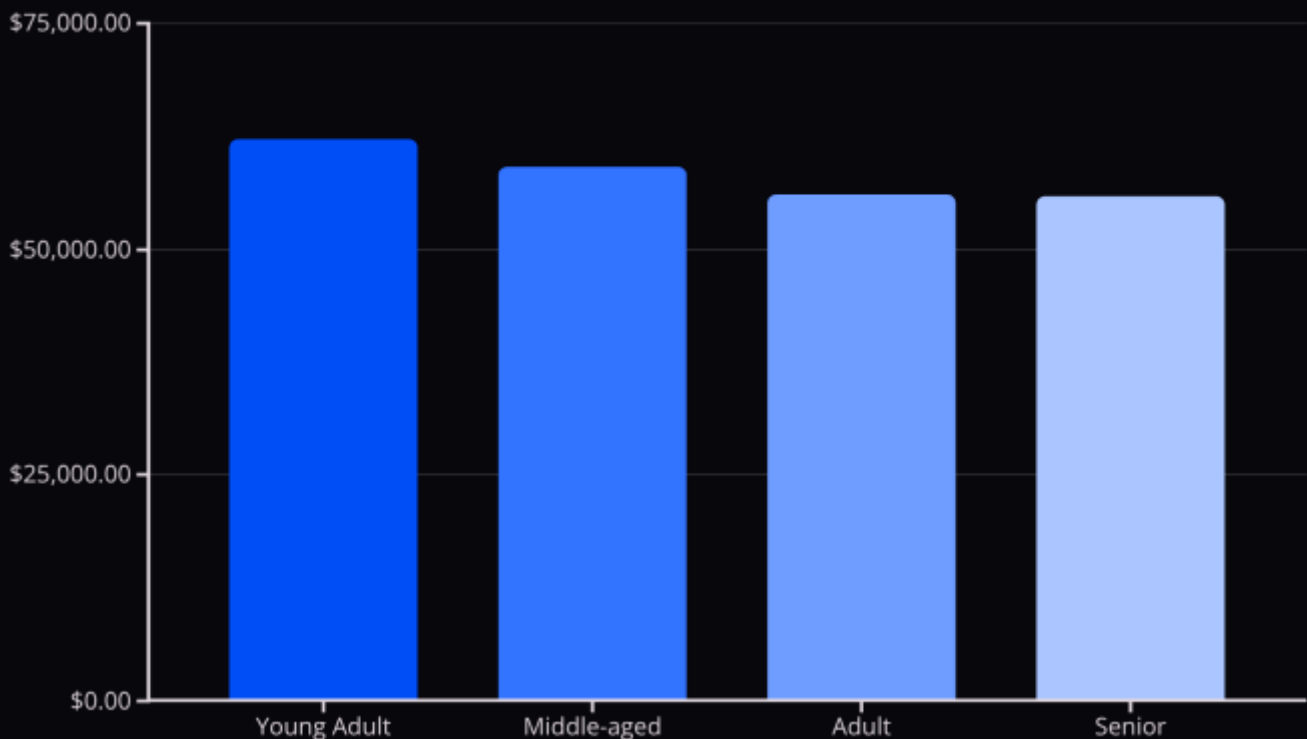
# Repeat Buyers & Revenue by Age

Repeat Buyers & Subscriptions



Repeat buyers are less likely to subscribe.

Revenue by Age Group



Young Adults contribute the most revenue.



# Power BI Dashboard: Visualizing Insights

Interactive dashboard for key metrics and visualizations.

## Key Metrics

3.9K Customers, \$59.76 Avg Purchase, 3.75 Avg Review.

## Visualizations

Subscription status, revenue by category/age, sales by category/age.

# Business Recommendations

## → Boost Subscriptions

Promote exclusive benefits.

## → Customer Loyalty Programs

Reward repeat buyers.

## → Review Discount Policy

Balance sales with margin control.

## → Product Positioning

Highlight top-rated and best-selling products.

## → Targeted Marketing

Focus on high-revenue age groups and express shipping users.